A Coin Flip Is Not a Red Light: Grading a Maternal-Health Risk Alarm for Mothers, Community Health Workers, and Clinicians

Maitreya Sapariya*
Independent Researcher

Aditi Patil
Smt. B. K. Shah Medical Institute & Research Centre, Sumandeep Vidyapeeth (Deemed to be University)

Abstract

A maternal-health risk model splits a case 0.487 against 0.499 between two adjacent risk levels. A clinician can read that margin and discount it; a mother handed a red alarm for the same prediction cannot. Rendering a coin flip as an emergency to the reader least able to interrogate it is the failure we address.

We render one prediction for three readers and gate the rendering on the model's own uncertainty: below a fixed top-two margin the highest-severity signal is withheld from the mother's channel and from no other. The derate is red to amber, never red to green, and amber still says *needs follow-up*.

Three claims, each bounded. Abstention rendered by stakeholder vulnerability rather than uniformly, which is an intersection of two established literatures and not the gating mechanism. Deterministic templating over SHAP attributions, with no language model at inference. And a leakage audit: retaining this benchmark's 561 duplicate rows moves macro-recall from 0.580 to 0.859, a measurement we report rather than a problem we claim to have found.

This is a design and feasibility artifact. Our evaluation is a formative single-evaluator critique of our own renderings: it found five defects and is not evidence that the tiers help anyone. Nobody in the three groups has used the system, and user studies are the next step, gated on ethics. The data is public, the task is generic maternal-health risk, and we make no clinical-validity claim.

Keywords: explainable AI; uncertainty; selective prediction; maternal health; community health workers; risk communication.

1 Introduction

A risk score computed at a rural antenatal visit is read by at least three people. A clinician wants the decomposition and can argue with it. A community health worker wants to know whether to refer this mother and what to write down. The mother wants to know what to do next, and she may not read at all, in which case the score reaches her as a colour and a spoken sentence or it does not reach her.

Most work on explaining a model to several audiences varies how much of the explanation each one gets. That is the right instinct, and it leaves one question open. When the model is not confident, what is each reader *entitled to be told*?

*Corresponding author. ORCID 0009-0003-9346-3775.

Consider a case from the dataset we use. The model puts it at 0.013, 0.487 and 0.499 across low, mid and high risk. The top two are separated by 0.012, and the model is wrong: the record is true mid-risk. A clinician shown that split can see the call was nearly a tie. The mother, at the end of the same pipeline, has no way to discount anything, and if the rendering follows the argmax alone she is the one who receives an emergency generated by a coin flip.

Our response is to make the abstention decision part of the rendering rather than part of the model. One band is computed per prediction from the class probabilities, and the highest-severity signal is raised only when the model is not on a knife edge. Below the threshold the mother's channel is derated while the clinician's is untouched. **The derate is red to amber, never red to green.** Amber still says *needs follow-up* and still sends her to a health worker. Nothing in this design removes a mother from care, and a reader who takes "not an emergency" to mean "not a concern" has read it backwards.

1.1 What we claim, and what we do not

We claim three things.

First, and this is the headline, **abstention differentiated by stakeholder vulnerability**. The decision to withhold is taken once, from the model's output, then rendered as different assertions for different readers, graded so that the reader with the least power to interrogate the system receives the most conservative signal.

Second, **deterministic templating over SHAP attributions**. Every string a health worker or a mother receives is a fixed template filled from the case's attributions. No language model runs at inference, so the rendering can be regenerated, diffed and audited, and there is nothing in it to hallucinate.

Third, **a quantified leakage audit** of a widely used benchmark. The UCI Maternal Health Risk dataset contains 561 exact duplicate rows. Scoring the same model under the same cross-validation with those rows retained moves macro-recall from 0.580 to 0.859, which lands inside the accuracy band the literature reports for this dataset.

The disclaimers belong in the same breath. **We do not claim that explanations should be audience-specific**, which is settled and was settled before us [1, 11, 13]. **We do not claim the gating mechanism.** Declining to answer under uncertainty has a long lineage in selective prediction and learning to defer, and our fixed-margin rule is a hand-set and cruder instance of it [9, 18]. **We do not claim a modelling contribution.** Fitting a classifier with SHAP attributions to this dataset is saturated [16, 17], and our model is a vehicle for the rendering question. On the audit we claim the measurement and not the discovery: the duplicates are the kind of thing noticed in a notebook and never written up, and we report what they cost rather than asserting we found them first.

1.2 What kind of evidence this is

This is a design and feasibility artifact, and we would rather say so here than let a reader discover it in Section 6. No mother, no community health worker and no clinician has used any of these renderings.

Our evaluation is a formative, single-evaluator heuristic critique conducted on our own artifact. It found five defects, including a driver list that could print a reading arguing against its own prediction under a heading announcing it as a flag, and all five were fixed before this paper was written. That is what the evaluation is: an instrument that improved the design. It is not evidence that any tier helps anyone, and the nearest empirical work argues against assuming so, since community health workers in India have been reported struggling with AI explanations even after simplification [21, 26]. Studies with the three groups are the next step, gated on ethics clearance. The data here is public, the task is generic maternal-health risk, and nothing in this paper predicts an outcome for a real patient.

Section 2 places the work against the two literatures it sits between, Section 3 covers the dataset and the leakage measurement, and Section 4 gives the rendering rule and the three tiers. Section 5 reports the evaluation defects first, Section 6 collects what the design cannot yet support, and Section 7 gives the commands that regenerate every artifact here.

2 Related Work

This work sits at the meeting point of two mature literatures, each of which has settled a question we depend on and neither of which has asked the question we ask. Each subsection below states what is already occupied.

2.1 Stakeholder-tailored explanation

That different audiences need different explanations of the same model output is established, not open. Arya et al. made the point in a title — *One Explanation Does Not Fit All* — and built the toolkit and taxonomy to go with it [1]. Imrie et al. show that stakeholders in healthcare impose genuinely divergent interpretability requirements, not differing degrees of one requirement [11]. Kim et al. build and compare stakeholder-tailored explanation interfaces directly, including the clinician-versus-patient split that two of our three tiers reproduce [13]. Bello et al. go further and propose a *three-level* pipeline from technical explanation to natural language, which is our structural twin [2]. **We therefore claim no novelty for the idea that explanations should be audience-specific, nor for tiering as an architecture.** Both predate us.

Our contrast with Bello et al. is one of mechanism rather than structure, and it is one of our stated contributions. Their mediator is a large language model. Ours is a fixed set of templates over SHAP attributions, so every string a health worker or a mother receives is determined by the case and can be regenerated and audited. Nothing is generated at inference time and there is nothing to hallucinate. We give up fluency and coverage for a rendering an auditor can check line by line.

Suresh et al. present the sharpest challenge to how we have organised this paper. They argue for moving *beyond* role labels, since a stakeholder's role determines neither the knowledge they bring nor the interpretability task they face, and they offer a typology that decouples the two [28]. Our tiers are named by role: clinician, ASHA, mother-to-be. We concede the point rather than wait for it to be made. Those names are shorthand, and what actually varies across the tiers is what each reader lacks, in reading modality, literacy and decision authority. Each tier is designed against that deficit. In this deployment the two track each other closely, since the ASHA is a specific, minimally-schooled cadre and the mother may not read at all. Szymanski et al. report that role and expertise co-vary in practice while remaining analytically separable [29], which supports role as a serviceable proxy here without rescuing it as a general axis. Where the two come apart, a deployment would have to select on the deficit.

2.2 Uncertainty communication, abstention, and selective prediction

The second literature is where our claim lives, so we are precise about its boundary. Withholding a prediction when the model is not confident is a long-settled mechanism. Hendrickx et al. survey machine learning with a reject option and name the case we implement: rejection under *ambiguity*, where two classes are close, as distinct from rejection under novelty [9]. Mozannar and Sontag give the learning-to-defer version, where the model is trained to route a case to a human expert rather than answer it [18]. Our rule declines the highest-severity signal when the top-two class probabilities are separated by less than a fixed margin, which is a hand-set instance of the same mechanism and a cruder one than either of these. **We claim none of it.** Kompa et al. supply the clinical framing: in medical machine learning, abstaining on high-uncertainty cases is a safety property rather than a shortfall in coverage [14]. That is the argument we borrow to justify derating a signal instead of issuing it.

Bhatt et al. are the nearest ancestor of our headline claim [4]. They treat uncertainty as itself a form of transparency and consider explicitly how it should be *communicated to different stakeholders*. Our delta is narrow: they make the *display* of uncertainty audience-aware, whereas we make the *suppression* of an assertion audience-aware. The mother does not receive the uncertainty rendered more gently, she receives a different assertion, while the clinician receives the split probabilities unmodified. The abstention decision is taken once, from the model's own output, then rendered per audience and graded so that the reader with the least power to interrogate the system receives the most conservative signal. We find no prior work that makes the abstention *rendering* a function of stakeholder vulnerability. That intersection, and only that intersection, is what we claim.

2.3 Explainable AI for community health workers in the Global South

Okolo et al.'s systematic review of explainable AI in the Global South records how little of this work reaches the people it describes: across the corpus they surveyed, deployment with actual users is close to absent [20]. That is the gap this paper sits in.

Three studies define our context and our principal exposure. Okolo et al. interviewed community health workers in rural India about AI-enabled mobile health tools and about explanations specifically [19, 21]. Solano-Kamaiko et al. built and studied an explorable explanation interface with the same cadre [26]. Srinidhi et al. document ASHA Kirana, a deployment in which ASHAs collect data that feeds an algorithm and returns a report for a doctor [27], which is our data flow exactly, minus the tiering and the uncertainty gate.

Two concessions follow, and both weaken us. First, this literature does not report that simplification suffices. Okolo et al. and Solano-Kamaiko et al. both find community health workers struggling to interpret AI explanations even when those explanations have been simplified [21, 26]. The comprehensibility of our ASHA card is therefore an open empirical question, and that finding is the clearest statement of why a user study is the necessary next step rather than an optional one. Second, every paper named in this subsection has users and this paper has none. We propose a rendering rule and evaluate the artifact that implements it, which is evidence of a weaker kind than theirs, and not a claim to know better than they do what a community health worker needs.

2.4 Communicating risk without numeracy

Our third tier renders to a reader who may not read, and the risk-communication literature is both its warrant and its sharpest critic. Galesic et al. show that icon arrays let low-numeracy readers extract risk information they otherwise miss [7], and García-Retamero and Cokely's systematic review finds visual aids improve risk literacy most among exactly the vulnerable readers our tier targets [8]. Richter et al. add the negative result that matters most for a voice channel: purely verbal communication of benefits and harms performs poorly for patients with limited health literacy [24], which is why our spoken message is paired with a visual rather than shipped alone. Peters et al. collect current best practice for numeric risk [22], and Bradley et al. test risk presentation in the maternity setting [5].

That literature recommends conveying *magnitude* transparently, through an icon array or a visible denominator. Our mother tier strips magnitude entirely and renders an ordinal band plus a single action. Read flatly, we are doing what the field advises against.

The resolution turns on a precondition those recommendations assume: an icon array communicates a magnitude the sender is entitled to assert. Our headline case is precisely the one where the model is not so entitled, at a top-two probability split of 0.487 against 0.499. Rendering that as roughly fifty chances in a hundred would convey a precision the model does not possess, to the reader least equipped to discount it. The departure is therefore justified *by* this literature rather than in spite of it: where the magnitude is trustworthy, show it; where the uncertainty gate has fired, an ordinal band plus an action is the honest rendering. That is Section 2.2's argument arriving from the health-communication side, and it remains an argument rather than a result, since whether the band is understood as intended is untested here.

2.5 Machine learning on the UCI Maternal Health Risk dataset

Supervised classification with SHAP attributions on this dataset is saturated. Mamun et al. [17], Maisoon et al. [16], Widyawati et al. [30], Rahman et al. [23], Maheswari et al. [15] and Hossen et al. [10] all report the same broad construction: tree ensembles, resampling for class imbalance, cross-validation, and SHAP for post-hoc attribution. **There is no machine learning contribution available here and we do not claim one.** Our model is a vehicle for the rendering question, and that framing is forced by the field rather than chosen out of modesty.

Maisoon et al. are our closest methodological neighbour and deserve explicit separation [16]. They use the same dataset with the same two-row outlier removal, tree ensembles with SMOTE, stratified five-fold cross-validation, SHAP, and the same low-resource framing. Everything upstream of the explanation is substantially their pipeline. What is ours lies downstream: deduplicating before splitting and reporting what that costs (Section 3), gating the rendering on the model's own uncertainty, and rendering the gated output three ways. Their attributions and Widyawati et al.'s independently rank blood sugar and systolic blood pressure at the top [16, 30], as ours do, which corroborates the attributions rather than adding a finding of ours.

3 Data and the Model Vehicle

3.1 Dataset

We use the UCI Maternal Health Risk dataset (repository id 863), released under CC BY 4.0. It contains 1014 records collected in rural Bangladesh, each with six features (age, systolic and diastolic blood pressure, blood sugar, body temperature and heart rate) and a three-level risk label: low, mid, high. It is public data and no other data is used anywhere in this work. No real patient record, de-identified or otherwise, appears in this paper or in the repository that accompanies it.

The dataset supports *generic maternal-health risk* and nothing narrower. It carries no label for any specific condition, and none of the laboratory, urinary or obstetric-history fields that would let one be inferred. We use the generic term throughout, in the manuscript, in the code and in every rendered artifact, and no tier names a condition.

The label is a risk stratification recorded alongside the readings, not a delivered outcome. Nothing in this paper should be read as a prediction of what happens to any patient, and that constraint binds the third tier hardest: a spoken message naming a diagnosis would assert something the data cannot support. What we demonstrate is an explanation design over a model fit to this public dataset.

3.2 Deduplication, and why it happens before the split

Two preprocessing steps are applied, both before any train/test split, and both chosen so that neither learns a statistic from the data.

First, two records report a heart rate of 7 beats per minute. These are physiologically impossible and are dropped, leaving 1012 rows. Second, the remaining set contains a large number of exactly repeated feature-and-label vectors: 561 of the 1012 rows are exact duplicates

Table 1: The same model and the same cross-validation, scored with exact duplicate rows dropped and retained. Mean ± standard deviation across five stratified folds.

Duplicates	n	Macro-recall	Accuracy	High-risk recall
Dropped	451	0.580 ± 0.027	0.641 ± 0.025	0.723 ± 0.054
Kept	1012	0.859 ± 0.026	0.854 ± 0.026	0.908 ± 0.031

of an earlier row. We drop them, leaving **451 distinct records**: 233 low, 106 mid and 112 high, an imbalance of about 2.2:1 between the largest and smallest class.

Both steps are fixed rules rather than fitted transformations, which is why performing them before the split introduces no leakage: nothing about the test fold informs what is dropped. Everything that *does* learn from data, meaning feature scaling and SMOTE where a configuration used it, is fitted inside each cross-validation training fold only, through an `imblearn` pipeline. We considered the alternative, which is to retain the duplicates under group-aware cross-validation and keep more rows. We rejected it on two grounds: it adds machinery a vehicle model does not need, and deduplication makes our number directly comparable to what a reader gets from a plain `drop_duplicates`.

Deduplication does not resolve every ambiguity in the data. Among the 451 distinct records there remain **35 feature vectors that carry more than one risk label**: identical six-value readings recorded against different levels of risk. No model, however specified, can separate rows it cannot distinguish. We retain these rows rather than adjudicating between them, and return to what they imply for the achievable ceiling in Section 6.

3.3 What the duplicates were worth

The duplicate rows cost accuracy its meaning. A duplicated row can appear in a training fold and in the test fold of the same split, so retaining them lets a model be scored partly on records it has already memorised. We measured that effect directly. Scoring the identical model under the identical five-fold cross-validation twice, changing only whether duplicates were dropped, gives Table 1 (`results/tables/p1_duplicate_leakage.csv`).

Retaining the duplicates moves accuracy from 0.641 to 0.854 and macro-recall from 0.580 to 0.859. The inflated figure sits inside the band typically reported for this dataset, and the published range extends higher still: Maisoon et al. report 83.74% and Mamun et al. 99.51% [16, 17]. Neither describes duplicate handling, both report $n = 1014$, and preprocessing on this dataset is generally given only as normalisation and the removal of inconsistencies.

We are careful about what this licenses us to say. Leakage is well documented: Kapoor and Narayanan set out its taxonomy and its role in the reproducibility crisis [12], Rosenblatt et al. show that leakage through repeated subjects inflates performance and that the effect is worst on small datasets, which is exactly our situation [25], and Bernett et al. give the practitioner's checklist [3]. The contribution type is established too: re-analysing an inflated published result to recover the true number is what Eltawil et al. and Young et al. do [6, 31]. **We do not claim to be the first to notice these duplicates.** We searched for prior documentation of them in this benchmark and found none, but absence of evidence is weak here: a `drop_duplicates` result is exactly the kind of thing noticed in a notebook and never written up. What we claim is the measurement: the published band on this dataset is recoverable only when identical rows are allowed to straddle the split, and after deduplication macro-recall on this vehicle is 0.580. It is reproducible in either direction from the committed table. We report deduplicated figures throughout and state the comparison here, up front, so that our lower headline number is legible as hygiene rather than as a weaker model.

3.4 The model is a vehicle

We compared logistic regression, random forest and XGBoost, each with and without in-fold SMOTE, under five-fold stratified cross-validation (`results/tables/p1_model_metrics.csv`). The primary metric is **macro-recall**, with macro-F1, per-class recall, one-versus-rest ROC-AUC and the confusion matrix as support. Macro-recall is the right primary here because the costs are asymmetric: failing to flag a high-risk mother is the error that matters, and headline accuracy on an imbalanced three-class problem conceals exactly that failure.

The final model is a **random forest with no resampling** (`n_estimators=300`, `random_state=42`). It ties the SMOTE-resampled random forest on macro-recall at 0.580. We broke that tie deliberately rather than by argmax: the unresampled model has tighter variance across folds (±0.027 versus ±0.040), the better high-risk recall (0.723), and no dependence on a synthetic-oversampling step. It also lets SHAP run on the six raw clinical features in their own units, which is what makes the ASHA and mother tiers expressible as templates over attributions at all. The selection is a constant in the code, and the script still prints the argmax winner so the tie stays visible.

We present that 0.580 as the premise of the paper rather than as a shortfall to be excused. The design question this work addresses, how an uncertain prediction should be rendered to readers of differing power, only exists because models on data like this are uncertain. A model that was right 99% of the time would not need the machinery we describe. The attributions themselves are stable and unremarkable: blood sugar is the leading global driver for every class, with systolic blood pressure second, and diastolic pressure and heart rate close to noise (`results/tables/shap_global_mean_abs.csv`). We say "blood sugar and systolic pressure" rather than "the blood-pressure features" throughout, because the data supports the first and not the second.

3.5 Maternal age

The dataset contains very low maternal ages, reflecting adolescent pregnancy in the source population: 95 of the 451 distinct records are under 18 and the minimum is 10, both written to `p1_cohort_summary.csv` by the preparation step. We retain rather than filter these rows, because excluding a group the system would encounter is itself a harm. Our designated failure case shows what that costs: a 13-year-old whose recorded body temperature of 101 °F drives a true-low record to a predicted-high output, with blood sugar and systolic pressure both arguing against the flip. The model is over-weighting a demographic feature, which is why final judgment stays with a human and why the rendering rules of Section 4 exist. All data here is public; no real patient or child record is exposed.

4 Three Tiers from One Prediction

4.1 What the alarm is, and what derating it means

By *alarm* we mean the highest-severity signal a tier is permitted to raise: the red lamp and the urgent spoken message for the mother, the HIGH header on the health worker's card. The rule this section describes withholds that signal when the model is close to a coin flip, and substitutes a middle signal that still says *needs follow-up*.

The direction of that substitution is the whole design, so we state it before anything else. **The derate is red to amber, never red to green.** A near-tie does not mean relax. It means the system is not entitled to the loudest thing it can say, and falls back to the one below it, which still sends the mother to a health worker. Nothing here removes a mother from care.

4.2 The band rule

Each rendering is driven by one band, computed once per prediction from the class probabilities and shared across all three tiers:

- GREEN if the predicted class is low risk.
- RED if the predicted class is high risk *and* the top-two probability margin is at least 0.15.
- AMBER otherwise — which covers every mid-risk prediction and every high-risk prediction that sits inside the margin.

The threshold of 0.15 is a design constant. We did not tune it, and no experiment in this paper supports that value over 0.10 or 0.20; we return to what would be needed to justify it in Section 6. Computing the band once matters as much as the threshold does. In the first version of the renderer the health worker's card banded on its own rule and the mother's lamp banded on this one, so a single prediction could carry two identities. Section 5 describes how that was caught.

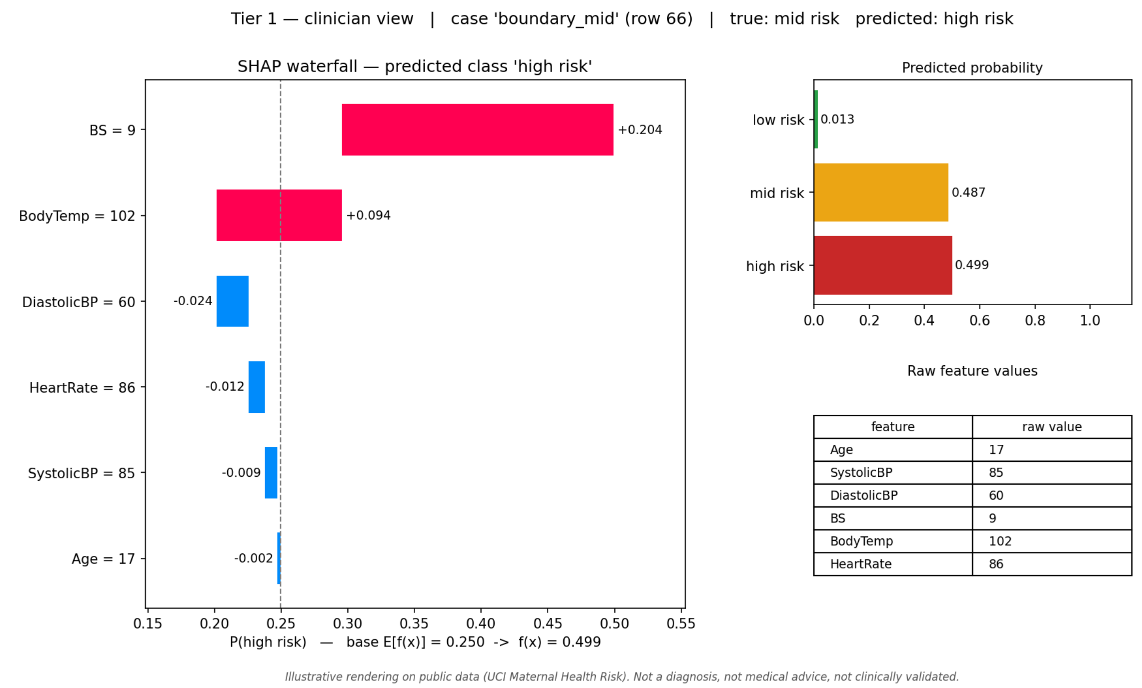
4.3 One prediction, three renderings

Figure 1 shows the same prediction rendered three ways. The case is row 66 of the deduplicated set, a true mid-risk record the model calls high risk, with class probabilities 0.013/0.487/0.499 across low, mid and high. The top-two margin is 0.012. Blood sugar contributes +0.204 to the predicted class and body temperature +0.094; the remaining four features contribute negatively and together amount to less than 0.05.

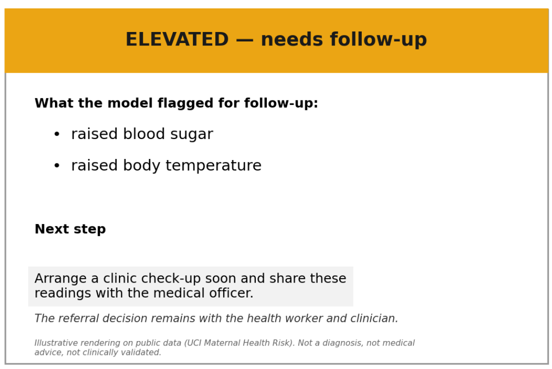
Clinician. The full decomposition, unmodified. Signed SHAP contributions per feature with numeric labels, all three class probabilities, and the six raw readings in clinical units. The 0.487 against 0.499 is shown as it stands. A reader who can interrogate the model is given what she needs to do so, including the fact that the call was nearly a tie.

Community health worker. A card carrying the band header, the factors behind *this* prediction, and one next step. Two rules shape it. The factors are local rather than global: naming the dataset's top features on every card would print the same two words for every mother and explain nothing about any of them. And only features that push *toward* the predicted class are listed, ranked by signed contribution, because a card headed "what the model flagged" that lists a reading arguing against its own prediction misleads the person making the referral. On this case the card reads ELEVATED — NEEDS FOLLOW-UP, lists raised blood sugar and raised body temperature, and asks the worker to arrange a check-up and share the readings. Every string is a fixed template constant filled from the attributions. No language model runs at inference, which is our stated alternative to LLM-mediated tiering [2]: the card is regenerable, diffable, and has nothing to hallucinate.

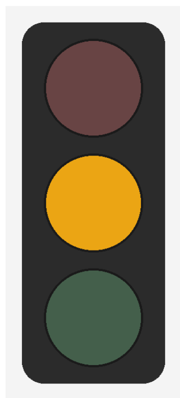
Mother-to-be. A three-lamp visual and one spoken Hindi sentence. No probability, no percentage, no condition named, and no text required to use it. The tier answers *what do I do next*, not *how likely is this*. Here the amber lamp lights and the message says, in Hindi, that some findings need attention, not to panic, and to get checked soon at her ANM or nearest health centre. The Hindi strings were written and clinically reviewed by the second author, a physician and native speaker; they are neither machine-translated nor author-invented.



(a) Clinician: signed SHAP contributions, all three class probabilities, and the raw readings.



(b) Community health worker: the band header, the two factors pushing toward the prediction, and one next step.



(c) Mother-to-be: amber lit, red dark.

Figure 1: Row 66 of the deduplicated set, rendered for all three readers. The record is true mid-risk; the model calls it high risk, at 0.013/0.487/0.499 across low, mid and high, a top-two margin of 0.012. The clinician panel carries that margin intact, 0.487 against 0.499, so the near-tie stays visible. The other two panels carry the derated band instead: the mother receives amber, and the red she would otherwise have received is withheld from her channel and from no other. Amber still tells her to get checked. All four cases across all three tiers are in the archived repository.

4.4 What we claim

Three literatures surround this design and we claim none of them. Tiering explanations by audience is established [1, 11, 13]. Withholding a prediction under uncertainty is a mechanism with a long lineage, and our margin rule is a hand-set instance of it [9, 18]. Fitting a classifier with SHAP attributions to this dataset is saturated [16, 17].

What we claim is the intersection, and it is narrow. Prior work on audience-specific uncertainty makes the *display* of uncertainty audience-aware [4]; the reader with less expertise receives the same uncertainty in a gentler presentation. Here the abstention decision is taken once and then rendered as *different assertions*. The clinician receives 0.487 against 0.499. The mother receives amber, and the red she would otherwise have received is withheld from her channel and from no other. The grading runs against power: the reader least able to interrogate the system gets the most conservative signal.

On this case the rule is doing what it was built for. The model is nearly split between mid and high, and it is wrong — the record is true mid-risk. The rendering that reaches the mother is the one that survives being wrong.

5 Formative Heuristic Evaluation

We rated the rendered artifacts against WCAG 2.1 and Nielsen's usability heuristics: 16 criteria across three tiers, 48 cells, each grounded in a named detail of a file on disk. Contrast ratios are computed at run time from the palette constants rather than typed in by hand.

This is a single-evaluator, author-conducted, formative assessment. It is not user testing and it is not validation. Nothing in it is evidence that the tiers help anyone. What it did was find defects in our own artifact, which is what we report.

5.1 What it found

The first pass ran against the original renderer and returned four failures, three of them on the health worker's card. All were fixed and the tiers re-rendered; the ratings below describe the corrected artifacts.

The driver list was sign-blind. Features were ranked by absolute SHAP magnitude with no sign filter, so a reading that argued *against* the prediction could be printed under a heading announcing it as a flag. On the false-positive case the card listed low blood sugar, whose contribution to the predicted class was -0.085. The routine card was worse: it listed the two features that were the model's reasons for saying *low* under that same heading, above a next step telling the reader to arrange a check-up. This is the defect that could have put a wrong cue in front of the person making a referral, and no model metric would have shown it.

The two tiers banded independently, so the confident high-risk case rendered an amber card beside a red lamp — one prediction with two identities. The band is now computed once and passed to both.

Header contrast was assumed rather than measured. White header text was hard-coded; on the amber header, which is the state for two of the four rendered cases, it measured 2.12:1. Ink is now chosen per band by measured contrast, worst case 5.14:1, and a self-check asserts it.

Direction words had no deadband, so a blood sugar of 7.7 against a median of 7.5 was worded as "raised". The word now attaches only outside 0.25 IQR of the median.

The mother's transcript carried no provenance, while both other tiers printed the disclaimer. It now records case, band and disclaimer under an explicit written-not-spoken rule; the spoken line is unchanged, because appending provenance to every message would bury the one action the tier exists to deliver.

5.2 What it could not settle

Across the 48 cells the distribution after those fixes is 29 pass, 12 partial, 1 fail and 6 deferred. The count is a description of coverage and not a score, and it is worth less than the five defects above. The single failure is the clinician tier shipping as an image of text. The six deferred cells are all keyboard operability and assistive-technology semantics, which a static PNG cannot exercise and a deployed application would.

Read by column, each tier fails where a single universal rendering would have to compromise: the clinician view needs chart and English literacy, the mother's view needs hearing, and neither gap closes inside its own tier without destroying what that tier is for. Read by row, the criteria are covered. The architecture is what conforms; no single output does.

None of this tells us whether a health worker or a mother understands what she is handed. That question needs users, and the nearest prior work is not encouraging about assuming the answer: studies with community health workers in India report difficulty interpreting AI explanations even after they have been simplified [21, 26]. Studies with mothers, health workers and clinicians are the next step, and they are gated on ethics clearance.

6 Limitations

6.1 Nobody has used this

No mother, health worker or clinician has seen any of these renderings. That is the limitation the rest depend on, and it bounds what the paper is allowed to conclude: we have a design and a formative critique of our own artifact, not evidence that the design works.

Four claims sit downstream of it. Whether a health worker reads the card as intended is assumed rather than shown, and the nearest evidence argues against assuming it, since community health workers in India have been reported struggling with AI explanations even after simplification [21, 26]. Whether withholding magnitude from the mother helps or harms her is an argument from the model's uncertainty, and Section 2.4 makes it, but the risk-communication literature it departs from is empirical and our departure is not. The 0.15 margin is a design constant with no experiment behind it, as Section 4 states. And role remains a proxy: we design against what each reader lacks in modality, literacy and decision authority, but we recruit nobody to check that the proxy holds, which is the substance of the objection Suresh et al. raise against role-based tiering [28].

All four resolve the same way. Studies with mothers, community health workers and clinicians, gated on ethics clearance, which this preprint does not have and does not claim.

6.2 What one public dataset can support

The work uses the UCI Maternal Health Risk dataset and nothing else. It is public, it carries generic three-level risk labels, and it names no clinical condition, so neither do we. Nothing here is clinically validated and nothing predicts an outcome for a real patient.

Two properties bound the vehicle. Macro-recall is 0.580 after deduplication, which we report as the honest figure rather than the inflated one (Section 3.3); and 35 of the 451 distinct records are feature vectors carrying more than one risk label, an ambiguity no model can resolve from these six features. That is a ceiling on the data, not a shortfall in the classifier.

One assumption inside the gate deserves naming. The margin rule thresholds a difference between random-forest class probabilities, and those are vote fractions rather than calibrated posteriors. We treat 0.15 as a design constant on an ordering, not as a statement about probability. Calibrating the model, and then asking whether the threshold should move, is future work.

6.3 Who the renderings still exclude

Each tier assumes something about its reader, and stating the assumption is what makes the gap findable.

The mother's visual carries no rendered text, so a deaf or hard-of-hearing mother gets nothing from it; the spoken sentence is her only channel and she cannot hear it. A rendered caption or an SMS companion would close this, and it reopens the literacy assumption the tier exists to avoid, so it needs user input rather than a unilateral design call. Colour is the lamp's primary cue. Position and the lit-versus-unlit luminance step are real redundancies and the spoken line is a second channel, but a reader who is both colour-blind and deaf is left with position alone, and position assumes the traffic-light convention. The clinician view needs chart literacy and English. The health worker's card is English-only, which is a scoped exposition choice rather than a claim about the architecture, since every string is a fixed template constant and localising it is substitution deferred to deployment. All three tiers ship as rasterised images of text.

6.4 The voice is a stand-in

Audio is synthesised with a general-purpose engine reached over the network. It is not tuned for Indian-language health speech, it handles acronyms unpredictably, and it needs connectivity that the setting this design describes may not have. Deployment-grade Indian-language synthesis (AI4Bharat, Bhashini) is the stated path. What we treat as the artifact is the Hindi source string, not the audio file.

6.5 Nothing is deployed

This is a design and feasibility artifact evaluated on a workstation. We make no field claim, no claim about performance under real clinic conditions, and no claim that the renderings have been carried to anyone by any delivery channel.

7 Reproducibility

Every figure, table and rendered artifact in this paper regenerates from five commands.

`python -m src.data` Fetches the dataset once from the UCI repository, caches it to `data/raw/`, and reports row count, duplicate count and missing values without modifying anything. Every later run reads the cache, so the rest of the pipeline needs no network.

`python -m src.model` Runs five-fold stratified cross-validation over three classifiers with and without SMOTE, fitting scaling and resampling inside each training fold. Writes `p1_model_metrics.csv`, `p1_duplicate_leakage.csv` and `p1_cohort_summary.csv`, which carries every cohort count this paper quotes. Saves the selected model.

`python -m src.explain` Computes global mean absolute SHAP per feature per class and local SHAP for the four criterion-selected cases. Writes three CSVs and five figures.

`python -m src.render` Renders the three tiers. It reads the saved SHAP tables rather than the model, so the renderer and the explainer cannot drift apart. `--case all` does all four cases.

`python -m src.evaluate` Rates the rendered artifacts across the 48 cells and writes `heuristic_matrix.csv` and `heuristic_evaluation.md`. Contrast ratios are computed from the renderer's own palette constants at run time.

7.1 Determinism

One seed, fixed at 42 in `src/config.py`, is applied to Python's hash randomisation, the standard library generator and NumPy by a `set_seeds()` call at the top of each script that touches randomness. The loader contains no random component and needs none. Dependencies are pinned to exact versions, and the dataset is cached after first download rather than re-fetched. Re-running the pipeline reproduces every PNG, CSV and text output byte for byte.

7.2 The exception

Speech synthesis is not ours to seed. The Hindi audio is produced by a remote service, and re-running the renderer on unchanged input returns files that are sometimes byte-identical and sometimes not. We state that as it is rather than rounding it to either extreme. The determinism guarantee above covers our pipeline and stops at that endpoint.

What is deterministic, and what we therefore treat as the artifact of record, is the Hindi source string. It lives in `results/tables/tier_mother_<case>_hi.txt`, it is committed, and it reproduces exactly. The MP3 is a rendering of that string.

One operational consequence is worth recording, because we were caught by it. Re-rendering over an existing audio file can leave the previous file in place while still reporting success and updating its timestamp. A cold run must empty `results/audio/` before re-rendering, or it can certify audio it did not regenerate.

7.3 Artifacts

The dataset is public and CC BY 4.0. Code, all rendered artifacts for four cases across three tiers, the cross-validation and SHAP tables, and the evaluation matrix are archived at `https://doi.org/10.5281/zenodo.22252076`, which is the concept DOI and resolves to the latest release; development history is at `https://github.com/stiFFLer-codes/three-voices`. The paper cites the committed tables directly, so any number in it can be checked against the file that produced it. The two-row leakage comparison in Table 1 is one of those files rather than a separate analysis: `p1_duplicate_leakage.csv` is written by the same command that writes the main metrics table, from the same cross-validation, so the comparison re-runs whenever the model does.

Data and Code Availability

The dataset is the UCI Maternal Health Risk dataset (repository id 863), public and released under CC BY 4.0, at `https://doi.org/10.24432/C5DP5D`. No other data is used and no real patient record appears in this work.

All code, the rendered artifacts for all four cases across all three tiers, the cross-validation and SHAP tables, and the evaluation matrix are archived at `https://doi.org/10.5281/zenodo.22252076`. That is the concept DOI and resolves to the latest release; development history is at `https://github.com/stiFFLer-codes/three-voices`. Every figure and table in this paper regenerates from the five commands listed in Section 7.

CRediT Author Contributions

Maitreya Sapariya — Conceptualization, Methodology, Software, Formal analysis, Writing – original draft.

Aditi Patil — Clinical domain expertise, Hindi authoring and review, ASHA-tier design input, Validation.

References

[1] Vijay Arya, Rachel K. E. Bellamy, Pin-Yu Chen, Amit Dhurandhar, Michael Hind, Samuel C. Hoffman, Stephanie Houde, Q. Vera Liao, Ronny Luss, Aleksandra Mojsilović, Sami Mourad, Pablo Pedemonte, Ramya Raghavendra, John Richards, Prasanna Sattigeri, Karthikeyan Shanmugam, Moninder Singh, Kush R. Varshney, Dennis Wei, and Yunfeng Zhang. One explanation does not fit all: A toolkit and taxonomy of ai explainability techniques, 2019. URL `https://arxiv.org/abs/1909.03012`.

[2] Marilyn Bello, Rafael Bello, María-Matilde García, Ann Nowé, Iván Sevillano-García, and Francisco Herrera. A three-level framework for llm-enhanced explainable ai: From technical explanations to natural language. *Information Systems Frontiers*, December 2025. ISSN 1572-9419. doi: 10.1007/s10796-025-10668-1. URL `http://dx.doi.org/10.1007/s10796-025-10668-1`.

[3] Judith Bernett, David B. Blumenthal, Dominik G. Grimm, Florian Haselbeck, Roman Joeres, Olga V. Kalinina, and Markus List. Guiding questions to avoid data leakage in biological machine learning applications. *Nature Methods*, 21(8):1444–1453, August 2024. ISSN 1548-7105. doi: 10.1038/s41592-024-02362-y. URL `http://dx.doi.org/10.1038/s41592-024-02362-y`.

[4] Umang Bhatt, Javier Antorán, Yunfeng Zhang, Q. Vera Liao, Prasanna Sattigeri, Riccardo Fogliato, Gabrielle Melançon, Ranganath Krishnan, Jason Stanley, Omesh Tickoo, Lama Nachman, Rumi Chunara, Madhulika Srikumar, Adrian Weller, and Alice Xiang. Uncertainty as a form of transparency: Measuring, communicating, and using uncertainty. In *Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society*, AIES '21, page 401–413. ACM, July 2021. doi: 10.1145/3461702.3462571. URL `http://dx.doi.org/10.1145/3461702.3462571`.

[5] Victoria Bradley, Alexandra Hunt, Eve Bunni, Carol Kingdon, Leila Finikarides, Laura Bonnett, Caroline Cunningham, Christy Burden, Jo Ficquet, William Parry-Smith, Elena Oteroromero, Arezou Rezvani, Mairead Black, Gemma Clayton, Gordon C. S. Smith, Deborah Lawlor, Jon Heron, and Abi and Merriel. How should we communicate information regarding birth choices to women?: An online randomised survey. *BJOG: An International Journal of Obstetrics & Gynaecology*, 132(13):2177–2185, July 2025. ISSN 1471-0528. doi: 10.1111/1471-0528.18314. URL `http://dx.doi.org/10.1111/1471-0528.18314`.

[6] Mohamed Eltawil, Laura Byham-Gray, Yuane Jia, Neil Mistry, James Parrott, and Suril Gohel. Comment on iacobescu et al. evaluating binary classifiers for cardiovascular disease prediction: Enhancing early diagnostic capabilities. j. cardiovasc. dev. dis. 2024, 11, 396. *Journal of Cardiovascular Development and Disease*, 13(1):46, January 2026. ISSN 2308-3425. doi: 10.3390/jcdd13010046. URL `http://dx.doi.org/10.3390/jcdd13010046`.

[7] Mirta Galesic, Rocio Garcia-Retamero, and Gerd Gigerenzer. Using icon arrays to communicate medical risks: Overcoming low numeracy. *Health Psychology*, 28(2):210–216, 2009. ISSN 0278-6133. doi: 10.1037/a0014474. URL `http://dx.doi.org/10.1037/a0014474`.

[8] Rocio Garcia-Retamero and Edward T. Cokely. Designing visual aids that promote risk literacy: A systematic review of health research and evidence-based design heuristics. *Human Factors: The Journal of the Human Factors and Ergonomics Society*, 59(4):582–627, February 2017. ISSN 1547-8181. doi: 10.1177/0018720817690634. URL `http://dx.doi.org/10.1177/0018720817690634`.

[9] Kilian Hendrickx, Lorenzo Perini, Dries Van der Plas, Wannes Meert, and Jesse Davis. Machine learning with a reject option: a survey. *Machine Learning*, 113(5):3073–3110,

March 2024. ISSN 1573-0565. doi: 10.1007/s10994-024-06534-x. URL http://dx.doi.org/10.1007/s10994-024-06534-x.

[10] Md. Sabbir Hossen, Pabon Shaha, Md. Saiduzzaman, Md. Shovon, Aklima Khatun Akhi, and Md. Sadiq Iqbal. An explainable ai driven machine learning approach for maternal health risk analysis. In *2024 27th International Conference on Computer and Information Technology (ICCIT)*, page 1135–1140. IEEE, December 2024. doi: 10.1109/iccit64611.2024.11022383. URL http://dx.doi.org/10.1109/iccit64611.2024.11022383.

[11] Fergus Imrie, Robert Davis, and Mihaela van der Schaar. Multiple stakeholders drive diverse interpretability requirements for machine learning in healthcare. *Nature Machine Intelligence*, 5(8):824–829, August 2023. ISSN 2522-5839. doi: 10.1038/s42256-023-00698-2. URL http://dx.doi.org/10.1038/s42256-023-00698-2.

[12] Sayash Kapoor and Arvind Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9):100804, September 2023. ISSN 2666-3899. doi: 10.1016/j.patter.2023.100804. URL http://dx.doi.org/10.1016/j.patter.2023.100804.

[13] Minjung Kim, Saebyeol Kim, Jinwoo Kim, Tae-Jin Song, and Yuyoung Kim. Do stakeholder needs differ? - designing stakeholder-tailored explainable artificial intelligence (xai) interfaces. *International Journal of Human-Computer Studies*, 181:103160, January 2024. ISSN 1071-5819. doi: 10.1016/j.ijhcs.2023.103160. URL http://dx.doi.org/10.1016/j.ijhcs.2023.103160.

[14] Benjamin Kompa, Jasper Snoek, and Andrew L. Beam. Second opinion needed: communicating uncertainty in medical machine learning. *npj Digital Medicine*, 4(1), January 2021. ISSN 2398-6352. doi: 10.1038/s41746-020-00367-3. URL http://dx.doi.org/10.1038/s41746-020-00367-3.

[15] B. Uma Maheswari, Aniket Dixit, and Alok Kumar Karn. Machine learning algorithm for maternal health risk classification with smote and explainable ai. In *2024 IEEE 9th International Conference for Convergence in Technology (I2CT)*, page 1–6. IEEE, April 2024. doi: 10.1109/i2ct61223.2024.10543709. URL http://dx.doi.org/10.1109/i2ct61223.2024.10543709.

[16] Manaal Maisoon, Sushma Pradeep, and M. R. Sai Chakith. Maternal health risk stratification in low-resource settings: Comparative analysis of machine learning models with xai. *IEEE Access*, 14:102905–102919, 2026. ISSN 2169-3536. doi: 10.1109/access.2026.3710610. URL http://dx.doi.org/10.1109/access.2026.3710610.

[17] Mohammad Mamun, Safiul Haque Chowdhury, Md. Omar Faruq, Mir Mohammad Azad, Bishakha Rani Biswas, Mohammed Ibrahim Hussain, and Muhammad Minoar Hossain. Identification of maternal health risk from optimal features using explainable machine learning. *Engineering Reports*, 7(11), November 2025. ISSN 2577-8196. doi: 10.1002/eng2.70491. URL http://dx.doi.org/10.1002/eng2.70491.

[18] Hussein Mozannar and David Sontag. Consistent estimators for learning to defer to an expert. In Hal Daumé III and Aarti Singh, editors, *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 7076–7087. PMLR, 2020. URL https://proceedings.mlr.press/v119/mozannar20b.html.

[19] Chinasa T. Okolo, Srujana Kamath, Nicola Dell, and Aditya Vashistha. "it cannot do all of my work": Community health worker perceptions of ai-enabled mobile health applications in rural india. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, CHI '21, page 1–20. ACM, May 2021. doi: 10.1145/3411764.3445420. URL http://dx.doi.org/10.1145/3411764.3445420.

[20] Chinasa T. Okolo, Nicola Dell, and Aditya Vashistha. Making ai explainable in the global south: A systematic review. In *ACM SIGCAS/SIGCHI Conference on Computing and Sustainable Societies (COMPASS)*, COMPASS '22, page 439–452. ACM, June 2022. doi: 10.1145/3530190.3534802. URL `http://dx.doi.org/10.1145/3530190.3534802`.

[21] Chinasa T. Okolo, Dhruv Agarwal, Nicola Dell, and Aditya Vashistha. "if it is easy to understand then it will have value": Examining perceptions of explainable ai with community health workers in rural india. *Proceedings of the ACM on Human-Computer Interaction*, 8(CSCW1):1–28, April 2024. ISSN 2573-0142. doi: 10.1145/3637348. URL `http://dx.doi.org/10.1145/3637348`.

[22] Ellen Peters, Paul K. J. Han, and Clara N. Lee. Communicating numeric risk information to patients. *Journal of General Internal Medicine*, 40(13):3228–3231, April 2025. ISSN 1525-1497. doi: 10.1007/s11606-025-09520-8. URL `http://dx.doi.org/10.1007/s11606-025-09520-8`.

[23] Anika Rahman and Md. Golam Rabiul Alam. Explainable ai based maternal health risk prediction using machine learning and deep learning. In *2023 IEEE World AI IoT Congress (AIIoT)*, page 0013–0018. IEEE, June 2023. doi: 10.1109/aiiot58121.2023.10174540. URL `http://dx.doi.org/10.1109/aiiot58121.2023.10174540`.

[24] Romy Richter, Jesse Jansen, Iris Bongaerts, Olga Damman, Jany Rademakers, and Trudy van der Weijden. Communication of benefits and harms in shared decision making with patients with limited health literacy: A systematic review of risk communication strategies. *Patient Education and Counseling*, 116:107944, November 2023. ISSN 0738-3991. doi: 10.1016/j.pec.2023.107944. URL `http://dx.doi.org/10.1016/j.pec.2023.107944`.

[25] Matthew Rosenblatt, Link Tejavibulya, Rongtao Jiang, Stephanie Noble, and Dustin Scheinost. Data leakage inflates prediction performance in connectome-based machine learning models. *Nature Communications*, 15(1), February 2024. ISSN 2041-1723. doi: 10.1038/s41467-024-46150-w. URL `http://dx.doi.org/10.1038/s41467-024-46150-w`.

[26] Ian René Solano-Kamaiko, Dibyendu Mishra, Nicola Dell, and Aditya Vashistha. Explorable explainable ai: Improving ai understanding for community health workers in india. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, CHI '24, page 1–21. ACM, May 2024. doi: 10.1145/3613904.3642733. URL `http://dx.doi.org/10.1145/3613904.3642733`.

[27] V Srinidhi, Baneen Karachiwala, Aditi Iyer, Bhavya Reddy, Vinalini Mathrani, Neha Madhiwalla, Vani Periodi, Anuradha Sreevathsa, Lakshmi Viswanatha, and Gita Sen. Asha kirana: when digital technology empowered front-line health workers. *BMJ Global Health*, 6(Suppl 5):e005039, September 2021. ISSN 2059-7908. doi: 10.1136/bmjgh-2021-005039. URL `http://dx.doi.org/10.1136/bmjgh-2021-005039`.

[28] Harini Suresh, Steven R. Gomez, Kevin K. Nam, and Arvind Satyanarayan. Beyond expertise and roles: A framework to characterize the stakeholders of interpretable machine learning and their needs. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, CHI '21, page 1–16. ACM, May 2021. doi: 10.1145/3411764.3445088. URL `http://dx.doi.org/10.1145/3411764.3445088`.

[29] Maxwell Szymanski, Vero Vanden Abeele, and Katrien Verbert. Disentangling stakeholder role and expertise in user-centered explainable ai. In *Proceedings of the 33rd ACM Conference on User Modeling, Adaptation and Personalization*, UMAP '25, page 32–39. ACM, June 2025. doi: 10.1145/3699682.3728351. URL `http://dx.doi.org/10.1145/3699682.3728351`.

[30] Lilik Widyawati and Neny Sulistianingsih. Explainable ensemble learning for maternal health risk in low-resource settings. *Jurnal RESTI (Rekayasa Sistem dan Teknologi Informasi)*, 9(5):1121–1130, October 2025. ISSN 2580-0760. doi: 10.29207/resti.v9i5.6765. URL `http://dx.doi.org/10.29207/resti.v9i5.6765`.

[31] Vanessa M. Young, Samantha Gates, Layla Y. Garcia, and Arash Salardini. Data leakage in deep learning for alzheimer's disease diagnosis: A scoping review of methodological rigor and performance inflation. *Diagnostics*, 15(18):2348, September 2025. ISSN 2075-4418. doi: 10.3390/diagnostics15182348. URL `http://dx.doi.org/10.3390/diagnostics15182348`.